

Rudra Sen Mallik

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Links: [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

Professional Summary

Detail-oriented Electronics and Communications Engineering graduate specializing in Web Development, Data Analysis, and Software QA. Proven experience building robust backend systems using Python (FastAPI) and PostgreSQL, alongside interactive frontends using ReactJS and Tailwind CSS. Adept at integrating WebSockets, NLP models, and automated workflows to deliver scalable software solutions.

Skills

- **Languages:** Python, JavaScript, TypeScript, SQL (PostgreSQL)
- **Backend Development:** FastAPI, Node.js, WebSockets, REST APIs, SMTP Workflows
- **Frontend Development:** ReactJS, Tailwind CSS, HTML5, CSS3, JavaScript Canvas
- **Data Analysis & AI/ML:** Natural Language Processing (NLP), Scikit-Learn, TF-IDF, Cosine Similarity
- **Tools & Platforms:** Git, GitHub, Linux, Postman

Professional Experience

Backend Developer Intern | CodeMischief Coding Solutions

Feb 2024 – Jun 2024

- Engineered robust Python FastAPI and PostgreSQL backend systems for enterprise-grade CRM/ERP systems, ensuring seamless integration with ReactJS frontends.
- Implemented automated SMTP workflows and dynamic PDF generation modules, improving documentation turnaround times.
- Designed custom PostgreSQL-based reporting queries to optimize data retrieval and visualization for client analysis.

Projects

Safety Monitoring Helmet for Industrial Workers | Final Year Capstone Project

- Developed an IoT-enabled safety helmet featuring real-time heart rate monitoring, fall detection, and RFID-based user authentication.
- Built the server architecture using FastAPI and PostgreSQL, implementing WebSockets to enable instant emergency alert notifications.
- Designed a responsive monitoring dashboard using ReactJS and Tailwind CSS for real-time data visualization.

Movie Recommendation System with NLP Analysis | NLP/AIML Project

- Built a recommendation engine using TF-IDF with bigrams over a comprehensive dataset of 35,000+ movies.
- Developed modular training and storage scripts utilizing Scikit-Learn, optimizing recommendations via normalized linear functions and Cosine Similarity.

Education

Bachelor of Technology in Electronics and Communications Engineering

MIT World Peace University, Pune, MH | 2021 – Present

- **CGPA:** 7.56/10
- **Relevant Coursework:** Data Structures and Algorithms (DSA), Database Management Systems (DBMS), Natural Language Processing, Artificial Intelligence and Machine Learning.

Leadership & Activities

- **Student Placement Representative (SOEEE)** | May 2023 – Jun 2024
- **Class Representative (1st Year Batch)** | Nov 2022 – Jun 2023